#uni #s26 #bu419

# Lecture 03: Insurability, Risk Classification, and Pricing

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## Economics of Risk Classifications

1. Separating consumers by risk type has the potential to produce large profits in the short run.
2. An insurer that fails to base premiums on information that is known will lose money due to adverse selection.
3. In a competitive market, differences in expected claims costs will produce differences in premiums across consumers.

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## Criteria for Good Risk Classification Variable

- **Homogeneity within classes:** Loss experience within each rate class is more similar to each other than to loss experience in other classes.
- **Reliable:** There should be a 1:1 relationship between the rating variable and loss experience, and this relationship should be relatively stable over time.
- **Separation between classes:** Every insured belongs to one and only one rate class.
- **Socially acceptable:** Insureds are willing to be categorized by the rate classes.
- **Provides incentives for loss control:** Variable captures action or behavior change of insured.

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## Characteristics of a Good Risk Classification Variable

- The CAS lists 16 criteria of a good risk classification variable.
- **Actuarial:** 4 characteristics related to the underlying losses themselves.
    1. Accuracy
        - Rating variable should be directly related to cost differences across insurance.
        - Most Important Criteria.
    2. Homogeneity:
        - All members of a group have the same expected claims costs.
    3. Credibility

        - Rating group should be large enough to measure cost with statistical accuracy.
    4. Reliability
        - Maintain stable cost differences over time.
- **Operational:** 5 characteristics related to how easy it is for insurers to classify according to the variable.
    1. Objective
        - The classes must be clearly defined, mutually exclusive, and exhaustive with little ambiguity.
    2. Administrative Cost
        - The cost of obtaining and verifying the information must be less than the value of the added accuracy.
    3. Ease of manipulation
        - It should not be easy for the insured to alter or fake the information.
    4. Intuitive relationship between variable and cost
        - There should be a logical connection to cost
    5. Minimize discontinuities
        - Extreme rate jumps between classes must be avoided.
- **Social:** 5 characteristics related to the acceptance of the variable by the insurers.
    1. Privacy
        - Insureds must not feel that the variable requires disclosing overly sensitive or invasive personal information.
    2. Causality
        - The variable should ideally have a direct causal relationship affecting loss frequency or severity.
    3. Controllability
        - Provides incentives for loss control by capturing actions or behaviors that the insured can change.
    4. Affordability
        - It should not unfairly restrict the affordability of coverage, particularly for high-risk individuals.
    5. Availability
        - It should not unfairly restrict the availability of coverage for high-risk individuals.
- **Legal:** 2 characteristics related to the legality of the use of the variable in practice.
    1. Allowable under regulations
        - The variable must be legally permissible to use within the specific jurisdiction.
    2. Not unfairly discriminatory

    - Premium differences must be firmly justified by demonstrable cost differences to be considered fair under the law.
    - Basically, the different premiums for each class should make sense to a common person.
  - *In the Zurich Insurance Co. v. Ontario case in 1992, the court encouraged the industry to find a new classification system that does not use age.*

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## Age as an Underwriting Variable

- Many jurisdictions are now eliminating age as an acceptable classification variable.
  - Automobile insurance uses age heavily.
    - Canadian statistics show traffic accidents are the leading cause of death for 15 to 19-year-olds.
    - Drivers over 75 have the highest fatality rate per kilometer driven.
    - However, it faces social backlash regarding fairness and affordability for young drivers.
  - On the other hand, life and individual health insurance also uses age, but it faces no public or regulatory pushback because the relationship between age, health risk, and mortality is universally accepted.
- Canadian insurers actively look at or use alternative variables to supplement or replace age.
  - Your licensed and driving record is used to assess experience rather than just chronological age.
  - Insurance scores (or Credit History) are highly predictive of risk but regulated differently by each province.
    - It is allowed and widely accepted in Quebec.
    - Banned for auto insurance in several other jurisdictions.
    - Recently introduced in Nova Scotia.
- Impact on auto insurance if age cannot be used to price insurance
  - Restrictions on risk classifications result in cross-subsidization. Low-risk drivers pay too much, and high-risk drivers pay too little.
  - Removing age as a variable in auto insurance would lead to more high-risk drivers on the road.

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## Ratemaking

- Ratemaking is the prospective process of establishing rates or premiums used in insurance.

- It's based on the fundamental insurance equation: **Money In = Money Out**
  - Money in refers to the premiums that the insurance company collects and the earned investment income on its capital.
  - Money out refers to the claims that the insurance company pays, as well as other expenses that it incurs.
- If Money In > Money Out, the company has a profit. Otherwise, they have a loss.
- In insurance, the simple pricing equation of **Price = Cost + Profit** is:

$$\text{Premium} = \text{Claims} + \text{Loss Adjustment Expenses (LAE)} + \text{Underwriting Expenses} + \text{Profit \& C}$$

- Objectives of Ratemaking:
  - Essential Objectives
    - Cover expected losses and expenses.
    - Produce rates that make adequate provision for contingencies
    - Encourage loss control.
    - Satisfy rate regulators
  - Non-essential but desirable objectives
    - Produce rates that are reasonably stable.
    - Be simple and easy to understand.
- Input Variables to Ratemaking:
  - Claim frequency (probability of an insured having a claim)

$$\text{Frequency} = \frac{\text{Number of Claim Counts}}{\text{Number of Exposures}}$$

  - Claim severity (average claim size)

$$\text{Severity} = \frac{\text{Total Amount of Claims}}{\text{Number of Claim Counts}}$$

  - Pure premium

$$\text{Pure Premium} = \frac{\text{Total Amount of Claims}}{\text{Number of Exposures}} = \text{Frequency} * \text{Severity}$$

- Overall Rate Adequacy is the need to ensure overall premium collected is enough to pay claims, LAE, Underwriting Expenses, and Profit and Contingencies.
  - Ensure that each group or individual pays premium proportional to their risk.
  - Not every risk, in short, has the same risk profile.
  - High-risk insurance should pay a higher premium.

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## Ratemaking in P&C Insurance

- Rate making is the process of establishing rates used in insurance or other risk transfer mechanisms.
- Two methods of rate making:
  1. Pure Premium Method
     - A method of rate making that calculates the rate needed to cover expected losses.

$$\text{Pure Premium} = \frac{\text{Losses}}{\text{Exposure Units}}$$

$$\text{Rate per Exposure Unit} = \frac{\text{Pure Premium} + \text{Fixed Expenses}}{1 - \text{Variable Expense Ratio} - \text{Profit \& Contingencies}}$$

  2. Loss Ratio Method
     - A method of rate making that adjusts current rates based on the ratio of actual losses to expected losses.

$$\text{Loss Ratio} = \frac{\text{Claims Paid}}{\text{Total Premium Earned}}$$

$$\text{Indicated Rate Change} = \frac{\text{Actual Loss Ratio}}{\text{Expected Loss Ratio}} - 1$$

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